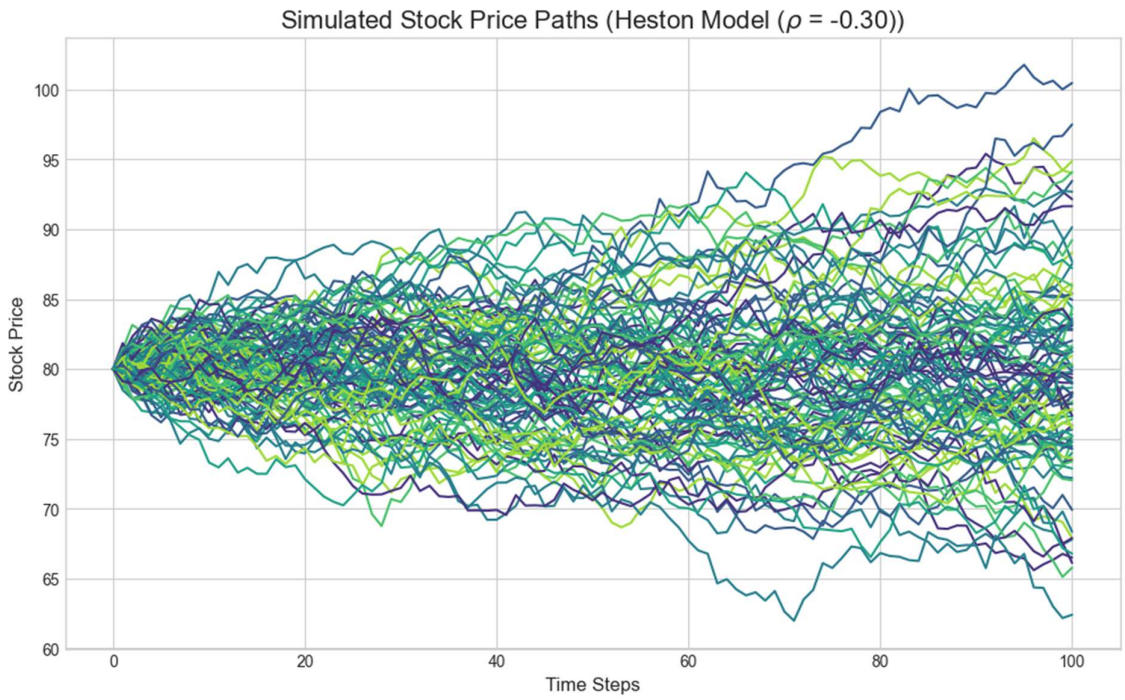
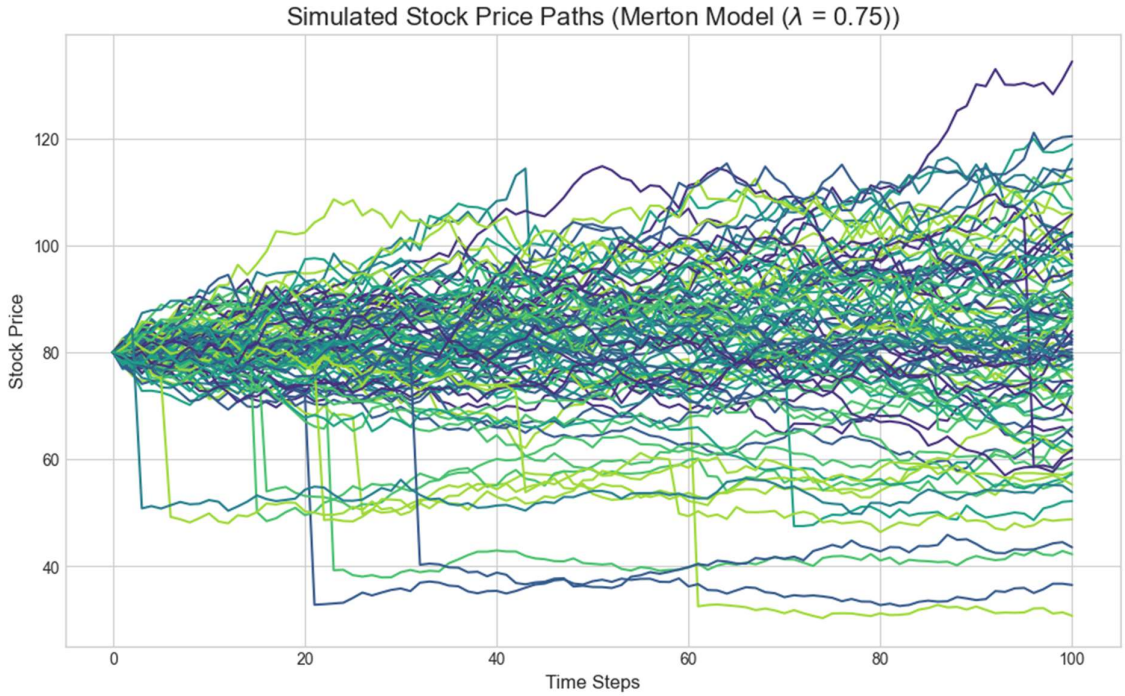
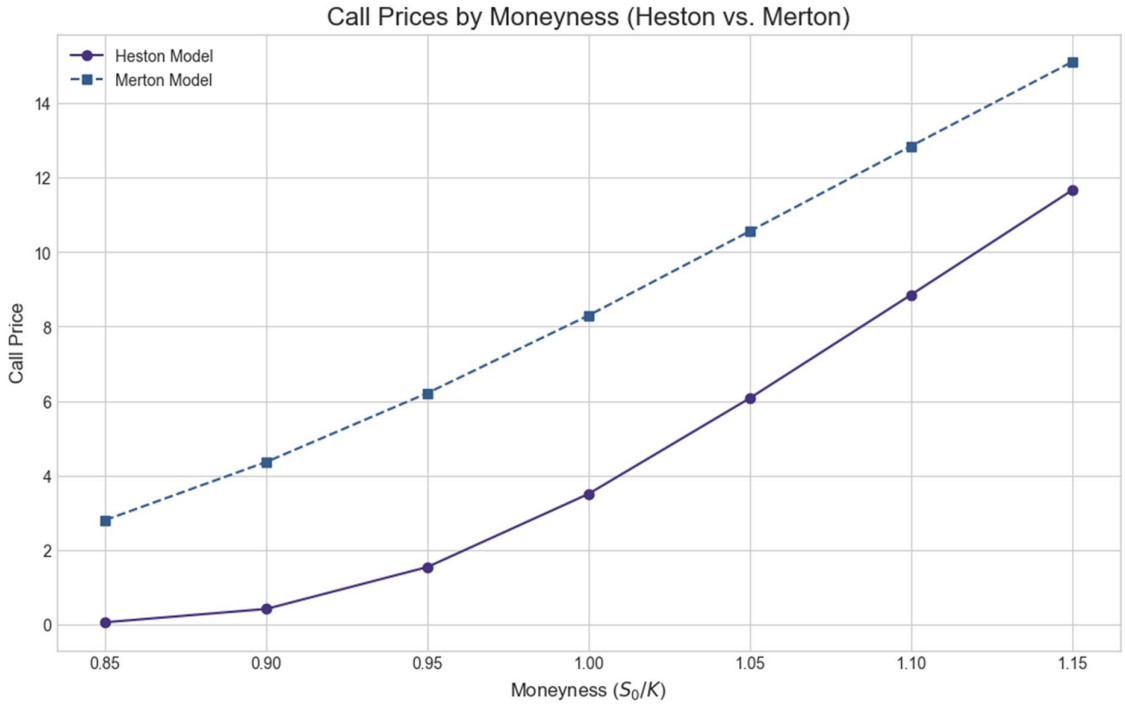
OUTPUT:

Generating Heston and Merton path plots...

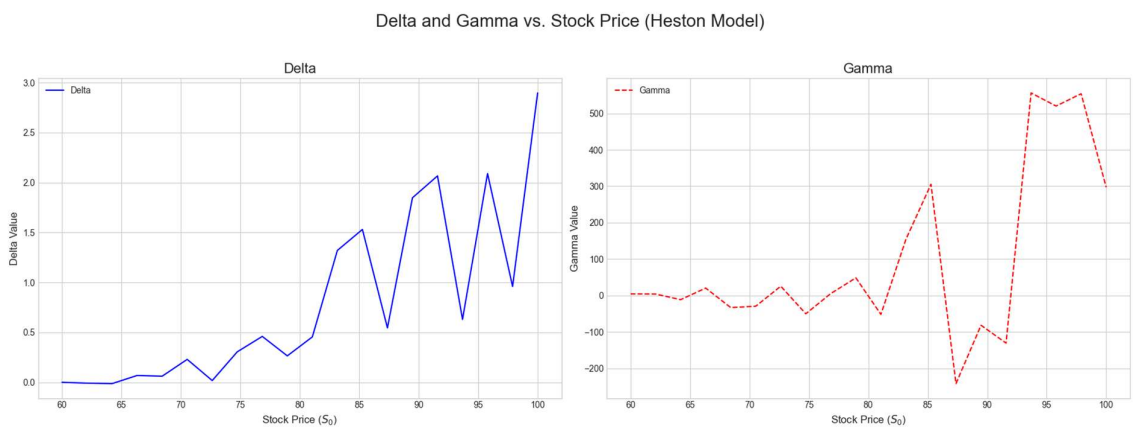




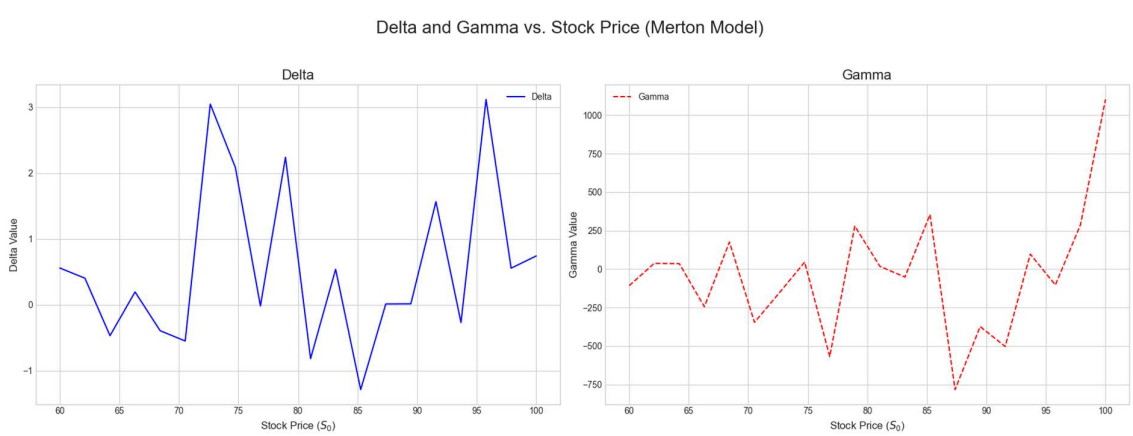
Generating Call Price vs. Moneyness plot...



Generating Greek vs. Stock Price plots for Heston...



Generating Greek vs. Stock Price plots for Merton...



All calculations and plots have been generated.

--- Technical Report Data ---

Calculations completed at: 2025-08-20 19:08:04

Heston Model Results (Q5, Q6, Q7)

Question	Correlation (ρ)	European Call Price	European Put Price
5	-0.3	3.49	2.39
6	-0.7	3.52	2.41

Question	Call Delta	Call Gamma	Put Delta	Put Gamma
7 (ρ=-0.30)	1.04	-342.93	-0.81	15.25
7 (ρ=-0.70)	0.56	-199.82	-0.82	23.79

Merton Model Results (Q8, Q9, Q10)

Question	Jump Intensity (λ)	European Call Price	European Put Price
8	0.75	8.29	7.23
9	0.25	6.8	5.73

Question	Call Delta	Call Gamma	Put Delta	Put Gamma

10 (λ=0.75)	-0.89	-1405.4	-1.1	701.94
10 (λ=0.25)	1.37	-181.25	1.98	577.28

Put-Call Parity Check (Q11)

Model	C - P (Actual)	S0 - K*exp(-rT) (Theory)
Heston (ρ=-0.30)	1.1	1.09
Heston (ρ=-0.70)	1.11	1.09
Merton (λ=0.75)	1.06	1.09
Merton (λ=0.25)	1.07	1.09

Call Prices by Moneyness (Q12)

Moneyness (K)	Heston Call Price	Merton Call Price
94.1176	0.06	2.8
88.8889	0.41	4.36
84.2105	1.54	6.21
80	3.5	8.3
76.1905	6.07	10.56
72.7273	8.85	12.84
69.5652	11.66	15.1

American vs. European Call and Greeks (Q13)

Option Type	Price	Delta	Gamma
European Call	3.48	0.79	-7.28
American Call	3.45	0.37	-118.35

European Up-and-In Call (Q14)

Option Type	Model	Strike	Barrier	Price
European Vanilla Call	Heston	95	N/A	0.04
European Up-and-In Call	Heston	95	95.0	1.37

European Down-and-In Put (Q15)

Option Type	Model	Strike	Barrier	Price
European Vanilla Put	Merton	65	N/A	2.76
European Down-and-In Put	Merton	65	65.0	10.77